



“India is a very close alternative to China for an OSAT hub, but an [Indian] OSAT ecosystem... requires strategic planning... [Else]... the Indian player may face trouble commercially”

DANISH FARUQUI
CEO, FAB ECONOMICS

Media Business, Americas, Tech Mahindra, explains his company’s role in the ecosystem. “We have been providing design services to several leading semiconductor companies globally for over a decade,” he says. This includes end-to-end design services in pre-silicon and post-silicon validation, embedded software, systems design, automated test engineering, and new-age solutions in IoT for various industry verticals. Tech Mahindra also provides turnkey system-on-chip (SoC) design services for new-generation silicon and derivative products in digital and analog domains.

BUT WHY OSATS?

OSATs are one of the key parts of the semiconductor supply chain. They are just like pure-play foundries but assemble, package, and test the wafers fabricated by any foundry (like TSMC) or integrated device manufacturer (IDM, like Intel). Setting up an OSAT is simpler and cheaper than a fab, and an OSAT can be up and running in a year and a half.

With chips becoming an integral part of everyday life and even being tested in the human brain, the semiconductor industry is expected to reach \$1 trillion by 2030. Given the growing demand for chips, many fabs are coming up worldwide.

Ajai Chowdhry, a co-founder of HCL, points out that only some of

the fabs coming up will have ATMP (assembly, testing, marking and packaging) plants sitting next to them. “So they will ship the wafer out to OSATs, who will do the packaging. Given that most of the packaging market is in Asia, that is where India can benefit,” says Chowdhry, the founder of Electronics Products Innovation Consortium. He is also on the advisory committee of the India Semiconductor Mission.

Sanjay Gupta, Chairperson of the India Electronics & Semiconductor Association (IESA), says, “Indian IT companies have expertise in software development, system integration, chip design, verification, and validation, which are essential for the semiconductor industry.”

Indian companies also have established partnerships and collaborations with global semiconductor players, such as Intel, Samsung, IBM, and TSMC, to provide them with various services and solutions, which is why they can leverage their existing capabilities and relationships to enter the OSAT and fab segments, he says.

RIGHT PLACE

Amongst the leading chip players, so far, only Micron Technology, a US-based memory chip maker, has announced plans to set up an ATMP plant in India. Indian companies building OSATs can bring multiple new opportunities against just one

THE OSAT LANDSCAPE

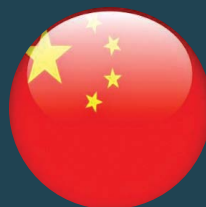
The Top 10 OSAT vendors by head-quarter location are distributed across Taiwan, China and the US

TAIWAN



**OSAT ASE,
PTI,
KYEC,
CHIPBOND,
CHIPMOS,
SIGURD**

CHINA



**JCET,
TFME,
HUATIAN**

THE US



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